Breakdown of Each Command

1. npm init -y

Word	Meaning
`npm`	Node Package Manager, the tool that installs and manages JavaScript packages. It comes bundled with Node.js.
`init`	Short for “initialize”. Tells npm to start a new project.
`-y`	Short for `--yes`. Answers “yes” to every question npm would normally ask (name, version, description, author, etc.) and uses the defaults.

What it does: Creates a `package.json` file in your folder. This file is your project’s ID card: it stores the project name, version, scripts, and the list of packages the project depends on.

Without `-y`: npm asks you questions one by one before creating the file.

2. npm install --save-dev typescript

Word	Meaning
`npm`	The package manager.
`install`	Download a package from the npm registry and put it in your project. Shortcut: `npm i`.
`--save-dev`	Record this package under `devDependencies` in `package.json`. This means “needed while developing, not needed when the finished app runs”. Shortcut: `-D`.
`typescript`	The name of the package to install. This is the TypeScript compiler.

What it does:

- Downloads TypeScript into the `node_modules/` folder.
- Adds `"typescript": "^x.x.x"` under `devDependencies` in `package.json`.
- Creates/updates `package-lock.json`, which locks exact versions.

Why dev dependency? The compiler is only needed to turn .ts into .js. The final JavaScript doesn’t need TypeScript to run.

3. npx tsc --init

Word	Meaning
`npx`	Runs a command from a locally installed package (in `node_modules/.bin`) without you having to install it globally. It comes with npm.
`tsc`	**T**ype**S**cript **C**ompiler. The program that converts .ts files to .js.
`--init`	A flag meaning “create a starter config file”.

What it does: Creates a `tsconfig.json` file. This file tells the compiler *how* to compile: which JavaScript version to output, where to put files (`outDir`), where to read from (`rootDir`), how strict to be, and so on.

4. npx tsc

Word	Meaning
`npx`	Runs the locally installed tool.
`tsc`	The TypeScript compiler.

What it does: With no file name given, `tsc` reads `tsconfig.json` and compiles **all** the .ts files it is configured to include.

Output (based on your config):

- `.js` files: the compiled JavaScript
- `.d.ts` files: type declarations (`"declaration": true`)
- `.js.map` files: link compiled JS back to the TS for debugging (`"sourceMap": true`)
- `.d.ts.map` files (`"declarationMap": true`)

These go into `dist/`, mirroring the folder structure of `src/`.

Compare: `npx tsc index.ts` compiles one file and **ignores** `tsconfig.json`. `npx tsc` alone uses the config.

5. node dist/Chapter_3_Setup_TS/index.js

Word	Meaning
`node`	The Node.js runtime. It executes JavaScript outside the browser.
`dist`	The output folder (short for “distribution”). It holds the compiled JavaScript.
`Chapter_3_Setup_TS`	A subfolder, mirrored from `src/Chapter_3_Setup_TS` because of `rootDir: "./src"`.
`index.js`	The compiled JavaScript file to execute.

What it does: Runs the compiled JavaScript file. Node cannot run .ts files directly, which is why you compile first.

Note: The `/` separates folders in the path (`dist` → `Chapter_3_Setup_TS` → `index.js`).

Summary Flow

```
npm init -y                        → creates package.json
        ↓
npm install --save-dev typescript  → installs the compiler
        ↓
npx tsc --init                     → creates tsconfig.json
        ↓
npx tsc                            → compiles src/*.ts into dist/*.js
        ↓
node dist/.../index.js             → runs the compiled code
```

Quick Reference: Terms

Term	Meaning
npm	Package manager for installing and managing packages
npx	Runner for locally installed package commands
tsc	TypeScript compiler
node	JavaScript runtime
`--flag`	Long-form option (e.g. --init, --save-dev)
`-f`	Short-form option (e.g. -y, -D)
package.json	Project info, scripts, dependencies
tsconfig.json	Compiler settings
node_modules	Folder where installed packages live
dist	Folder for compiled output

One-Line Meanings

Command	Meaning
`npm init -y`	Creates a `package.json` file with default answers (no questions asked).
`npm install --save-dev typescript`	Installs the TypeScript compiler as a development-only dependency.
`npx tsc --init`	Creates a `tsconfig.json` file with starter compiler settings.
`npx tsc`	Compiles all .ts files into .js (plus .d.ts and map files) using `tsconfig.json`.
`node dist/Chapter_3_Setup_TS/index.js`	Runs the compiled JavaScript file using Node.js.

To run one file quickly: `npx tsx src/Chapter_3_Setup_TS/index.ts`